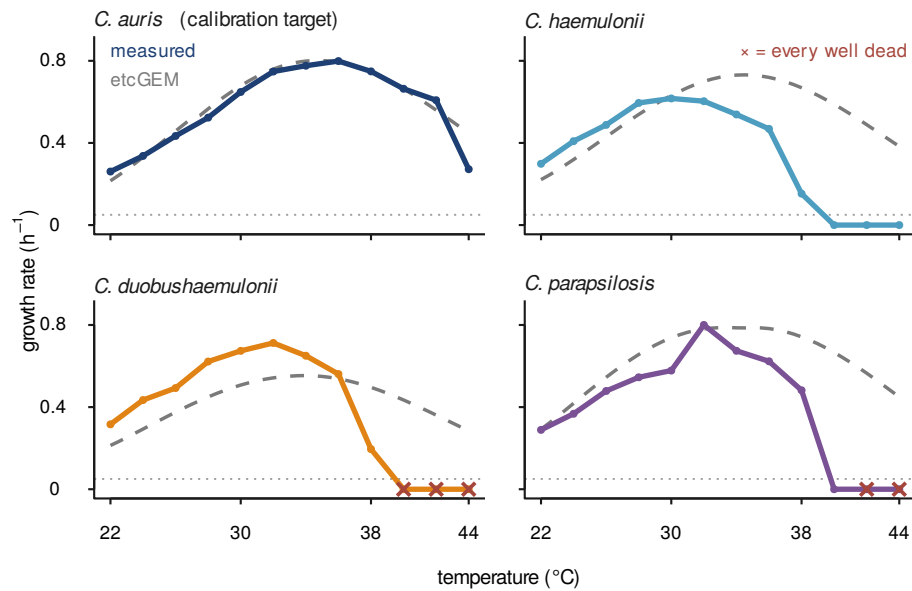
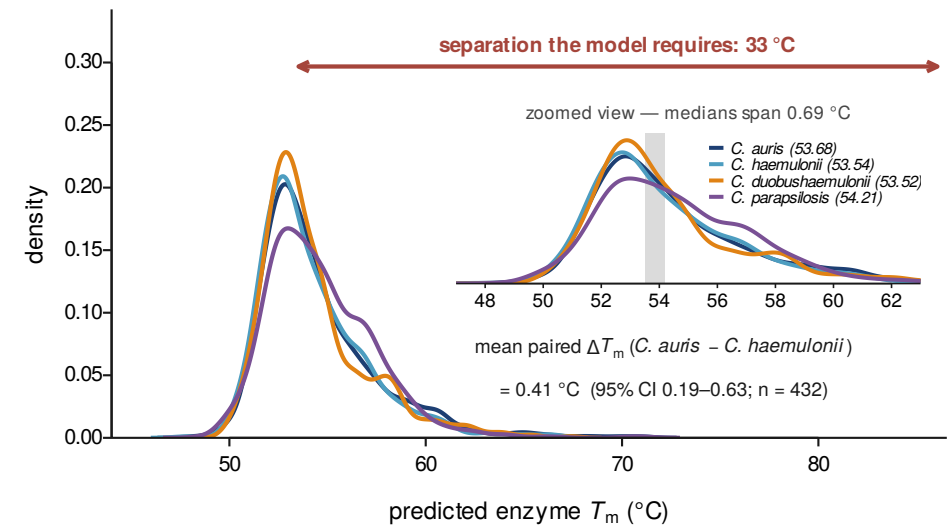


Within the etcGEM, sequence-predicted enzyme thermal properties do not reproduce the observed thermal divergence

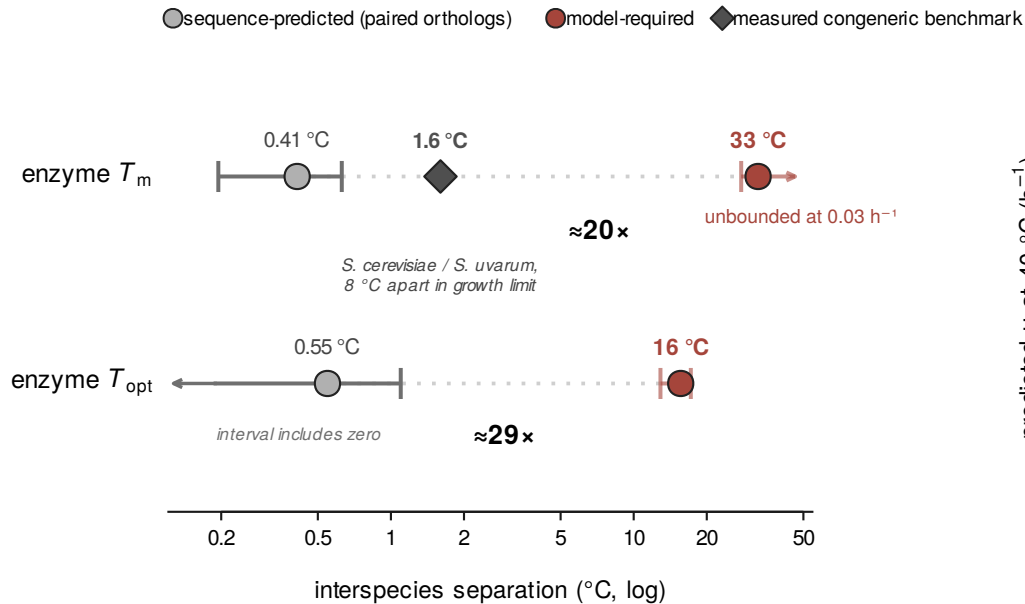
A The model fits its calibration target but misses the relatives' collapse



B Paired ΔT_m is directionally consistent but $\sim 79\times$ below the fitted requirement



C Required shifts exceed predictions and a measured benchmark



D Targeted perturbations do not drive relatives below detection

